

DATASCI 223 Final Project

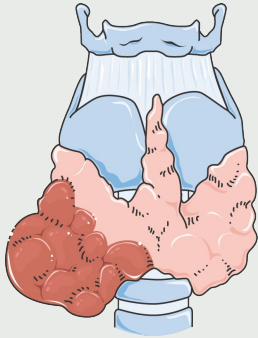
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Problem Statement

How accurately can machine learning models classify thyroid nodules as benign or malignant using patient clinical data?

- Lots of new and emerging research in the area of applying machine learning to predict whether an individual's thyroid tumor is benign or malignant based on patterns in clinical features in patient data [3]
- This motivates our research question: How accurately can machine learning models classify thyroid nodules as benign or malignant using patient clinical data?

Intro



- Over 40,000 cases of thyroid cancer are found each year [1].
- Most thyroid cancers do not produce noticeable signs or symptoms in the early stages of the disease [2].
- Symptoms include individual lump in the neck, change to one's voice, and trouble swallowing [2].

- Each year over 40,000 cases of thyroid cancer are diagnosed in the United States. [1]
- Thyroid cancer is a disease characterized by the uncontrolled growth of cells originating in the thyroid gland.

Intro cont.

- For this project, we will run different machine learning models, on the dataset
- Models include, logistic regression, neural networks, random forest, and XGBoost
- Models will be evaluated based on error metrics of accuracy, precision, recall, F1 score, and AUC

- Most cases are treated with surgery and have a 5-year relative survival of 98.5% [1]. Recent advances in artificial intelligence have attempted to apply machine learning to predict whether an individual's thyroid tumor is benign or malignant based on patterns in clinical features (e.g., free triiodothyronine (T3) level, gender) in patient data [3].
- Overall, machine learning has the potential to reduce unnecessary biopsies [5], unnecessary surgeries [6], and improve personalized treatments for thyroid cancer patients [7].

Description of the dataset

- The estimated dataset is simulated from real-world thyroid cancer-risk factors.
- Dataset contains around 20,000 rows and 17 columns
- Specifically examines certain ethnic groups, nationalities, and risk factors such as radiation exposure, iodine deficiency, and a family history of the condition

- The dataset is simulated from real-world thyroid cancer risk factors and comprises 212,691 rows (212,691 unique patients) and 17 columns (recorded patient clinical features). This dataset specifically examines certain ethnic groups, nationalities, and individuals with risk factors such as radiation exposure, iodine deficiency, and a family history of the condition.

How to run the code

- The code is a series of Jupyter notebooks, one for each mode. Different packages may be needed for each model
- Workflow:
 - Run “pip install -r requirements.txt”
 - data_cleaning_exploration/data_cleaning.ipynb
 - data_cleaning_exploration/exploratory_analysis.ipynb
- The following is up to the user's interest:
 - Random Forest: audrey_model.ipynb
 - XGBoost: chrissie_model.ipynb
 - Logistic Regression: emma_model.ipynb
 - Neural Networks: tara_model.ipynb, audrey_regular_nn_model.ipynb

- The code is a series of Jupyter notebooks. Two of the notebooks correspond to Exploratory Data Analysis and to preparing the data for machine learning
- There also exists one notebook for each model
- Audrey did random forest and neural network
- Chrissie did XGBoost model
- Emma did logistic regression model
- Tara did the feed forward network model

How to run the code cont.

- Packages include:
 - Numpy, Random, Pandas, Sklearn, Tensorflow, Matplotlib
- Model Versions:
 - Numpy (v.1.24.0)
 - Pandas(v.2.0.0)
 - Matplotlib(v.3.7.0)
 - Scikit-learn(v.1.3.0)
 - Tensorflow(v.2.15.0)
 - Jupyter(v.1.15.0)

- These are some of the packages and the version used
- State the packages and versions

Decisions

- Patient ID was removed from the list of predictors because it is arbitrary
- Thyroid Cancer Risk was removed from the list of predictors because it would create a data leakage and potential for circulatory analysis
- The training and test split was done with 70/30 split
- 5-fold cross-validation was used to tune hyperparameters of each model.

- The outcome we wanted to predict was diagnosis. Patient ID and Thyroid Cancer Risk were excluded from our predictors.
- Patient ID was removed because it is a variable that assigns a unique identifier to each patient, and does not offer useful information to help us decide what a patient's thyroid tumor diagnosis is. Thyroid Cancer Risk was removed because it would create a data leakage and potential for circulatory analysis, as Thyroid Cancer Risk describes very similar information to a patient's thyroid tumor diagnosis
- We applied a 70/30 training/test split
- On the training set, we applied 5-fold cross validation to tune the hyperparameters of each model
- We compared average 5 fold AUC on the training set to select the best set of hyperparameters for each model
- The optimized models were then applied on the test set, error metrics then calculated for the optimized models
- Model with highest AUC on test set was selected as final best model

Decisions cont.

- The following predictors from the original dataset were used: Age, Gender, Country, Ethnicity, Family History of Thyroid Cancer, Previous Radiation Exposure, Iodine Deficiency, Smoking, Obesity, Diabetes, TSH Level, T3 Level, T4 Level, Nodule Size, T3, and T4. Risk Factor Score, TSH Nodule Interact, and T3 TSH Ratio were predictors created and included in our analysis.

- Talk about the predictors in the original dataset ex) Smoking is a binary yes/no variable corresponding to whether the patient smokes or not; TSH Level, T3 Level, and T4 Level are variables corresponding to lab-measured hormone levels

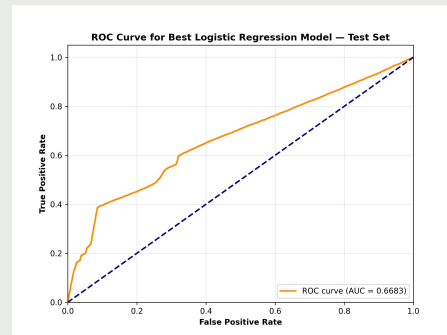
Decisions cont.

- Feature engineering was applied to existing columns in our data to transform raw data into more meaningful insights
- Three additional features were created through feature engineering
 - Risk Factor Score
 - TSH Level and Thyroid Nodule Size Interaction
 - Free T3 to TSH Level Ratio

- Three additional features were created through feature engineering: Risk Factor Score; TSH Level and Thyroid Nodule Size Interaction; and Free T3 to TSH Level Ratio

Example Output: Logistic Regression

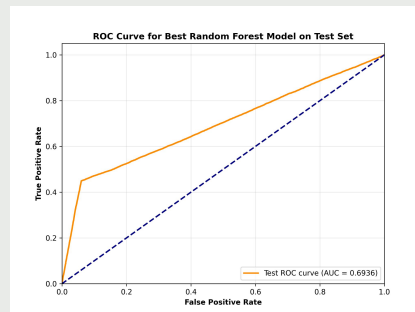
Accuracy	0.7803
Precision	0.6905
Recall	0.1004
F1	0.1753
AUC	0.6683



- Mention that these tables/graphs represent each model's performance on test set
- State the error metrics for the model
- Note that recall and F1 score are very low for this model → bad because we want to maximize true positives given goal of classifying thyroid tumors

Example Output: Random Forest

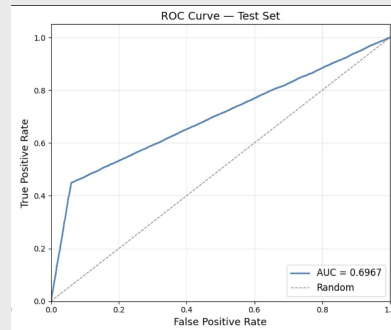
Accuracy	0.8256
Precision	0.6945
Recall	0.4469
F1	0.5439
AUC	0.6936



- State the error metrics for the model
- Note that recall is fairly low for this model

Example Output: Neural Network (Feed Forward)

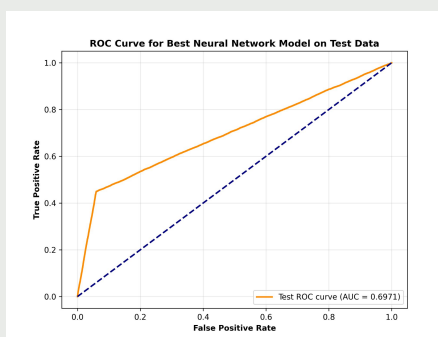
Accuracy	0.8259
Precision	0.6946
Recall	0.4485
F1	0.5451
AUC	0.6967



- State the error metrics for the model
- Note that recall is fairly low for this model

Example Output: Neural Network

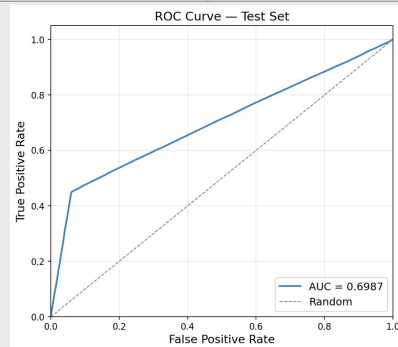
Accuracy	0.8260
Precision	0.6950
Recall	0.4483
F1	0.5451
AUC	0.6951



- State the error metrics for the model
- Note that recall is fairly low for this model

Example Output: XGBoost

Accuracy	0.8238
Precision	0.6949
Recall	0.4322
F1	0.5330
AUC	0.6987



- State the error metrics for the model
- Note that recall is fairly low for this model

Table Comparing All Models

	Logistic Regression	Random Forest	Neural Network (Feed Forward)	Neural Network	XGBoost
Accuracy	0.7803	0.8256	0.8259	0.8259	0.8238
Precision	0.6905	0.6945	0.6946	0.6952	0.6949
Recall	0.1004	0.4469	0.4485	0.4478	0.4322
F1	0.1753	0.5439	0.5451	0.5447	0.5330
AUC	0.6683	0.6936	0.6967	0.6971	0.6987

- This is the table comparing the error metrics for the four models created
- A bit disappointing that AUC's across models on the test set were not very high (all below 0.7)

Discussion

- We selected **XGBoost** as the top model because it had the highest AUC (0.6987)
- Used AUC as the primary selection metric because it measures how well a model discriminates between malignant and benign cases

- Based on the 5 models' metrics, we selected XGBoost as the top model because it had the highest AUC (0.6987). We note that the AUCs for the 5 models are fairly similar (range: 0.6683-0.6987). A future avenue is to explore more about why this occurred.

Benefits of XGBoost

- Handles both numeric and categorical variables very well
- Highly resistant to overfitting due to its built-in regularization parameters
- Capture complex and non-linear relationships between features that simpler models, such as logistic regression may miss

- One benefit of using an XGBoost model is that it is widely-known to perform well on structured, tabular data, which can be valuable, as we worked with such data in our analysis. XGBoost is also highly resistant to overfitting due to its built-in regularization parameters. It can also capture complex, non-linear relationships between features that simpler models, such as logistic regression, may miss. Finally, XGBoost is highly computationally efficient, which is crucial for analyzing large datasets like ours.

Limitations of XGBoost

- Difficult to explain the results of our analyses to clients and patients because of its limited interpretability
- Since the highest AUC was only 0.6987, there is definitely room for improvement and that perhaps the model needs to be changed in order for it to be more reliable
- Class imbalance: struggle to correctly classify underrepresented classes when the dataset is disproportionately skewed toward benign or malignant cases

- It can be difficult to explain the results of our analyses to clients and patients because of their limited interpretability
- The highest AUC was only 0.6987, indicating that there is definitely room for improvement and that the model may need to be revised to be more reliable
- class imbalance is another limitation, as XGBoost may struggle to correctly classify underrepresented classes when the dataset is disproportionately skewed toward benign or malignant cases.

Limitations of XGBoost cont.

- Model demonstrated low precision, recall, and F1 scores, indicating that it often fails to correctly identify true malignant cases
- This is very concerning in the medical context and can have very serious consequences for patient outcomes

- The model also demonstrated low precision, recall, and F1 scores, indicating that it often fails to correctly identify true positive malignant cases. This is very concerning in the medical context and can have very serious consequences for patient outcomes.

Issues Encountered

- No major issues were encountered
- Since data was simulated, it lacked the imperfections associated with clinical data
- Model performance not exceptionally high
- Further tuning of hyperparameters or alternative predictors could be used to improve the discrimination properties

- No major issues were encountered, because data was simulated it lacked the imperfections associated with clinical data. Despite this, model performance was not exceptionally high. Models struggled to have recall over 0.5, and no model had AUC over 0.7. This is problematic, as it suggests that our models make lots of false negatives and that our models only distinguish between classes modestly well. Further tuning of hyperparameters or alternative predictors could be used to improve the discrimination properties.

Contributions

Each student worked on an Individual Model. Students worked on data cleaning, documentation, and slides together.

Logistic Regression: Emma Chang

XGBoost: Chrissie Lok

Random Forest: Audrey Su

Feed Forward Neural Network: Tara Hosseini

Neural Network: Audrey Su

- Each student worked on individual model. Students worked on data cleaning, documentation, and slides together
- Each student states which model they worked on

Citations

- [1] Boucai, L., Zafereo, M., & Cabanillas, M. E. (2024). Thyroid Cancer: A Review. *JAMA*, 331(5), 425–435. <https://doi.org/10.1001/jama.2023.26348>
- [2] Mayo Foundation for Medical Education and Research. (2025, December 23). *Thyroid cancer*. Mayo Clinic. <https://www.mayoclinic.org/diseases-conditions/thyroid-cancer/symptoms-causes/syc-20354161>
- [3] Ma, F., Yu, F., Lv, S., Zhang, L., Lu, Z., Zhou, Q., Mao, H. R., Zhang, L., & Xiang, N. (2025). Machine learning model for differentiating malignant from benign thyroid nodules based on the thyroid function data. *BMJ open*, 15(5), e093466. <https://doi.org/10.1136/bmjopen-2024-093466>
- [4] Xiao, D., Wu, X., Xie, P., Lai, B., Zhong, J., Zhong, J., & Zeng, X. (2026). Study on differentiating benign and malignant thyroid nodules based on CT multi-phase artificial intelligence models. *Frontiers in endocrinology*, 16, 1738342. <https://doi.org/10.3389/fendo.2025.1738342>
- [5] Chen, L., Wang, Y., Jing, H., Bao, R., Sun, B., Zhang, M., & Luo, Y. (2025). Ultrasound radiomics models improve preoperative diagnosis and reduce unnecessary biopsies in indeterminate thyroid nodules. *Frontiers in endocrinology*, 16, 1615304. <https://doi.org/10.3389/fendo.2025.1615304>
- [6] Grégoire D'Andréa, Jocelyn Gal, Loïc Mandine, Olivier Dassonville, Clair Vandersteen, Nicolas Guevara, Laurent Castillo, Gilles Poissonnet, Dorian Culié, Roxane Elaldi, Jérôme Sarini, Anne Decotte, Claire Renaud, Sébastien Vergez, Renaud Schiappa, Emmanuel Chamorey, Yann Château, Alexandre Bozec, Application of machine learning methods to guide patient management by predicting the risk of malignancy of Bethesda III-V thyroid nodules, *European Journal of Endocrinology*, Volume 188, Issue 3, March 2023, Pages 249–257, <https://doi.org/10.1093/ejendo/lvad017>

- Above are the citations that we used for our work on the Datasci 223 Final Project

Citations

- [7] Pozdeyev, N., White, S. L., Bell, C. C., Haugen, B. R., & Thomas, J. (2026). Artificial Intelligence Applications in Thyroid Cancer Care. *The Journal of clinical endocrinology and metabolism*, 111(2), 316–324. <https://doi.org/10.1210/clinem/dgaf530>
- [8] *Thyroid hormone: What it is & function*. Cleveland Clinic. (2026, February 24). <https://my.clevelandclinic.org/health/articles/22391-thyroid-hormone>
- [9] *TSH (thyroid-stimulating hormone) levels: Symptoms*. Cleveland Clinic. (2026b, February 24). <https://my.clevelandclinic.org/health/articles/23524-thyroid-stimulating-hormone-tsh-levels>
- [10] *Triiodothyronine (T3) test: What it is, Function & Levels*. Cleveland Clinic. (2026b, March 9). <https://my.clevelandclinic.org/health/diagnostics/22425-triiodothyronine-t3>
- [11] *T4 (thyroxine) test: What it is, function, Purpose & Levels*. Cleveland Clinic. (2026a, February 24). <https://my.clevelandclinic.org/health/diagnostics/24235-thyroxine-t4-test>